Name :- Chethnika Sri

Register NO! - 192511112

Course code :- CSA1306

CO3 - ASSIGNMENT

DESIGN AND APPLICATION OF CONTEXT-FREE GRAMMAR FOR PROGRAMMING LANGUAGE SYNTAX ANALYSIS

TASK 1 - CONTEXT-FREE GRAMMAR DESIGN AND DERIVATION

CONTEXT-FREE GRAMMAR

A context-free Grammar (CFG) is used to define the syntax of programming languages

The production rules are:

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

Grammar Components

Start symbol : E

Terminals : {id, +, *, (,)}

Non-terminals : {E, T, F}

Explanation

E represents an expression involving addition

T represents a term involving multiplication

F represents an identifier or parenthesized expression.

Multiplication has higher precedence than addition. Because multiplication is represented at the T level.

4. Leftmost Derivation of a+b*c

In leftmost derivation, the leftmost non-terminals is replaced first

$E \Rightarrow E+T$
$\Rightarrow T+T$
$\Rightarrow F+T$
$\Rightarrow a+T$
$\Rightarrow a+T*F$
$\Rightarrow a+F*F$
$\Rightarrow a+b*F$
$\Rightarrow a+b*c$

5. Rightmost derivation of a+b*c

In rightmost derivation, the rightmost non-terminal is replaced first.

$E \Rightarrow E+T$
$\Rightarrow E+T*F$
$\Rightarrow E+T*c$
$\Rightarrow E+F*c$
$\Rightarrow E+b*c$
$\Rightarrow T+b*c$
$\Rightarrow F+b*c$
$\Rightarrow a+b*c$

6. Parse tree



The tree shows that b*c is evaluated before addition CFG defines valid arithmetic expression and ensures correct operator precedence.

Task 2 - Simplification of Context-Free Grammar

Given Grammar

$S \rightarrow AB \mid a$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid C$

$C \rightarrow C$

1) Identify Nullable Variable

A variable is nullable if it can produce ε

Given $A \rightarrow \varepsilon$

Therefore:

Nullable Variable - {A}

2. Elimination of ε - production

Since A is nullable, remove A from $S \to AB$

New productions

$S \to AB|B|a$

$A \to bB|C$

$C \to c$

The ε production is eliminated.

3. Elimination of unit Productions

A unit production has only one Variable on the right hand side.

Given:

$B \to C$ Replace it with $C \to c$

Also, $S \to B$ is a unit production Replace it with production of B $S \to AB|bB|c|a$

4. Removal of Useless Symbols

A useless symbol is one of that cannot generate a terminal string or cannot be reached from start symbol

- A, B and S are useful
- C is no longer required because its production has been replaced.
- Therefore remove C

FINAL SIMPLIFIED CFG

$S \rightarrow AB \mid bB \mid Ca$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid c$

Results:-

No ε-productions

No unit production

No useless symbol

The grammar is simplified by eliminating nullable, unit and useless production.

TASK 3- CONVERSION INTO CHOMSKY NORMAL FORM

Given grammar:

$S \rightarrow aAB \mid b$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

Definition:-

Chomsky Normal form (CNF) is a form of CFG in which every production must be in one of these forms

$A \rightarrow BC$

$A \rightarrow a$

Where B and C are variables and a is a terminal.

2. IDENTIFY INVALID PRODUCTION

The following productions do not satisfy CNF

$S \rightarrow aAB$ (contains 3 symbols and a terminal)

$A \rightarrow aA$ (contains a terminal and a Variable)

$B \rightarrow bB$ (contains a terminal and a Variable)

3. INTRODUCE NEW Variable:-

$X \rightarrow a$

$Y \rightarrow b$

4. Convert longer production

FOR $S \rightarrow aAB$:

$S \rightarrow XAB$

Introduce another Variable

$D \rightarrow AB$

Therefore $S \rightarrow XD$

for $A \rightarrow aA$:

$A \rightarrow XA$

for $B \rightarrow bB$:

$B \rightarrow YB$

5 FINAL CNF Grammar

$S \rightarrow XD \mid b$

$D \rightarrow AB$

$A \rightarrow XA \mid a$

$B \rightarrow YB \mid b$

$X \rightarrow a$

$Y \rightarrow b$

Verification

Production	CNF FORM
$S \rightarrow XD$	$A \rightarrow BC$
$S \rightarrow b$	$A \rightarrow a$
$D \rightarrow AB$	$A \rightarrow BC$
$A \rightarrow XA$	$A \rightarrow BC$
$A \rightarrow a$	$A \rightarrow a$
$B \rightarrow YB$	$A \rightarrow BC$
$B \rightarrow b$	$A \rightarrow a$
$X \rightarrow a$	$A \rightarrow a$
$Y \rightarrow b$	$A \rightarrow a$

All productions satisfy CNF rules

The given CNF is converted into Chomsky Normal form using New variable and binary productions

TASK 4 - Conversion into Greibach normal form (GNF)

Given Grammar

$S \rightarrow AB \mid a$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid b$

Definition

Greibach normal form (GNF) is a form of CF in which every production starts with a terminal symbol followed by zero or more variables

General form

$$A \rightarrow a\alpha$$

Where a is terminal and α is sequence of variable

2. Identify invalid production

$$S \rightarrow AB$$

Begins with a variable A not satisfy GNF

3. APPLY Grammar Transformation

$$A \rightarrow aA \mid a$$
$$S \rightarrow aAB \mid aB$$

The original production $S \rightarrow a$ is also retained.

4. FINAL GNF Grammar

$$S \rightarrow aAB \mid aB \mid a$$
$$A \rightarrow bB \mid b$$
$$B \rightarrow bB \mid b$$

Verification:

Production	First Symbol
$S \to aAB$	Terminal a
$S \to aB$	Terminal a
$S \to a$	Terminal a
$A \to aA$	Terminal a
$A \to a$	Terminal a
$B \to bB$	Terminal b
$B \to b$	Terminal b

Derivation of String aab

$$S \Rightarrow aAB$$
$$\Rightarrow aaB$$
$$\Rightarrow aab$$

The grammar is converted into GNF, and the string aab is successfully generated.

Task 5 - Context-Free Language Properties and Pumping Lemma

Given G language:

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

Introduction:

The pumping lemma is used to prove whether certain language are not context-free.

The given language contains equals number of a's, b's and c's in that order.

2 Context - Free Pumping Lemma

For every context - free language, there exists a pumping length p such that any string $|w|$ with $|w| \geq p$ can be divided into five parts

$$w = uvxyz$$

Subjects to the following conditions

1) $|vxy| \leq p$
2) $|vy| \geq 1$
3) $uv^i xy^i z \in L$ for all $i \geq 0$

3. Select a Suitable String

Choose

$w = a^p b^p c^p$ Clearly w belongs to L

4. Apply pumping conditions

Since $|vxy| \leq p$ the substring vxy cannot cover all three blocks of a's, b's and c's

Also $|vy| \geq 1$ so at least one symbol is repeated or removed.

5. Contradiction

Take $i = 0$ (remove v and y)

After pumping down:

The number of a's and b's or c's changes

At least one of the three symbol counts becomes different

Therefore, the resulting string cannot have equal number of a's, b's and c's

Hence:

$$uv^0xy^0z \notin L$$

This contradicts the pumping lemma condition

Therefore, the assumption that L is context free is false

Conclusion:-

$L = \{a^n b^n c^n \mid n \geq 1\}$ is not a context-free language (CFL)

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